Operations Management

Introduction to Operations Management & Process Analysis

SPIT- Minor in Management

28th August 2023

Session 1 and 2: Agenda

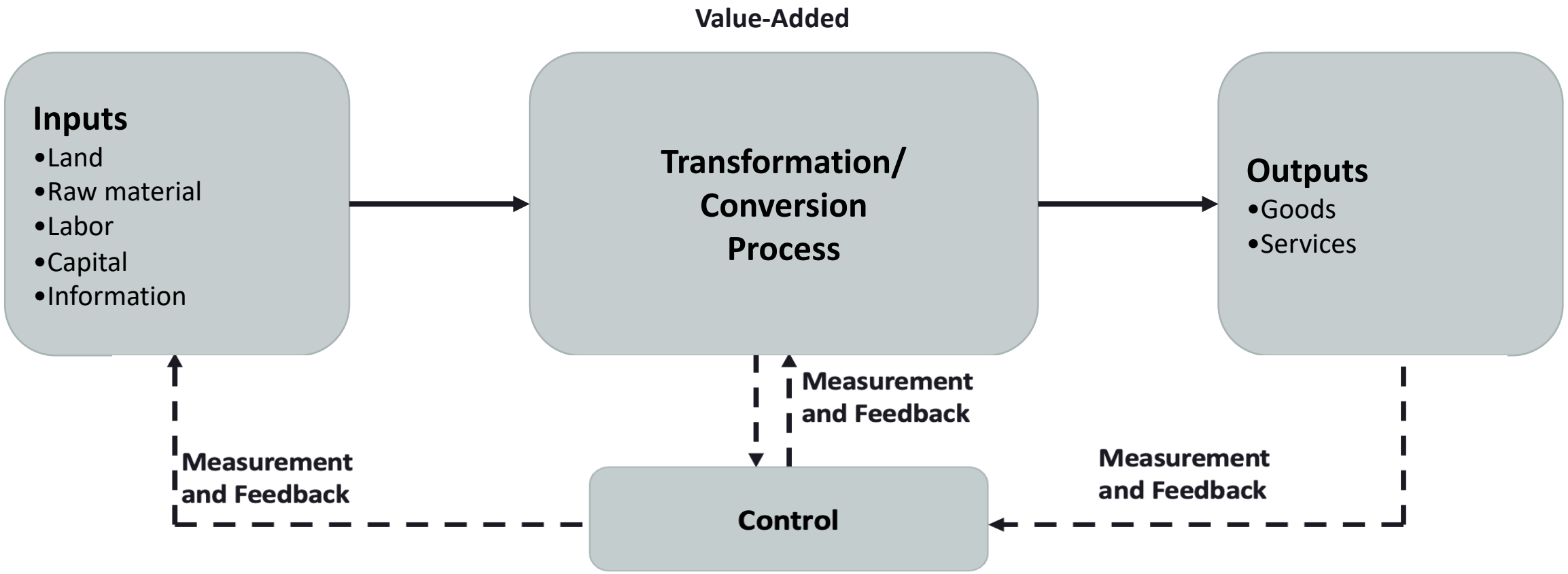
01 **Introduction to Operations Management**

02 Process Analysis

Operations Management

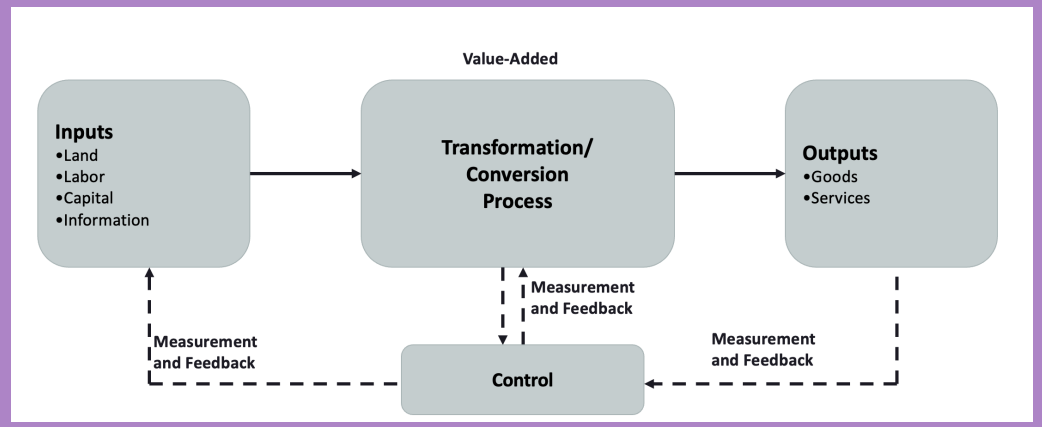
- **Operations** refers to *manufacturing and service processes* used to *transform inputs into outputs* desired by customers.
- **Operations Management is** about *designing, managing, and improving* the activities involved in *creating and delivering goods and services* within a budget while *meeting the requirements of customers*.
  - **Goods** are physical items produced by business organizations
  - **Services** are activities that provide some combination of time, location, form and have psychological value
- The key objective in operations is to **match supply and demand**

OM: The Transformational Process



Source: Operations Management, William J. Stevenson, 12th Edition, ISBN 978-0-07-802410-8, page 5.

Operations Objective : Matching Supply and Demand



Supply

Company

≈

Demand

Customer

Why do we need to balance Demand and Supply?

- Excess supply can prove to be wasteful and cost incurring
- Excess demand can lead to lost opportunity, loss of revenue, and possible customer dissatisfaction.

Matching Supply and Demand for products and services

	Manufacturing: Consumer Durables (Mobile Phone)	Services: Air travel
Supply	Units of mobile phone produced	Seats on a specific flight
Demand	New purchase from customers	Travel requirement on a specific time between origin and destination
Supply exceeds Demand	Inventory holding costs Money held in inventory	Empty seats
Demand exceeds Supply	Profit not realized, Dissatisfied customer	Overbooking Customer shift to alternate flights and modes
Possible Action	Better forecasting, response	Dynamic pricing

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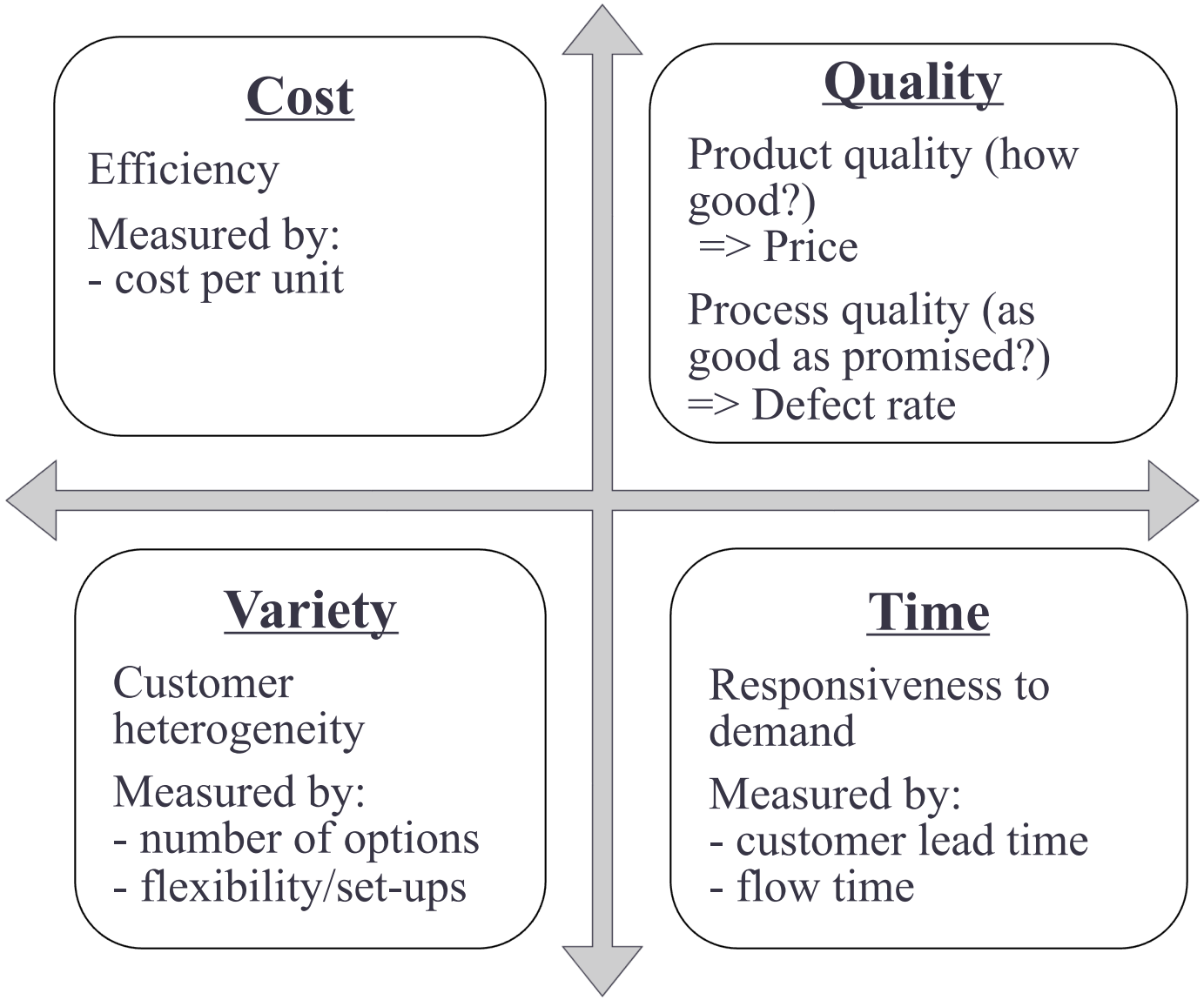
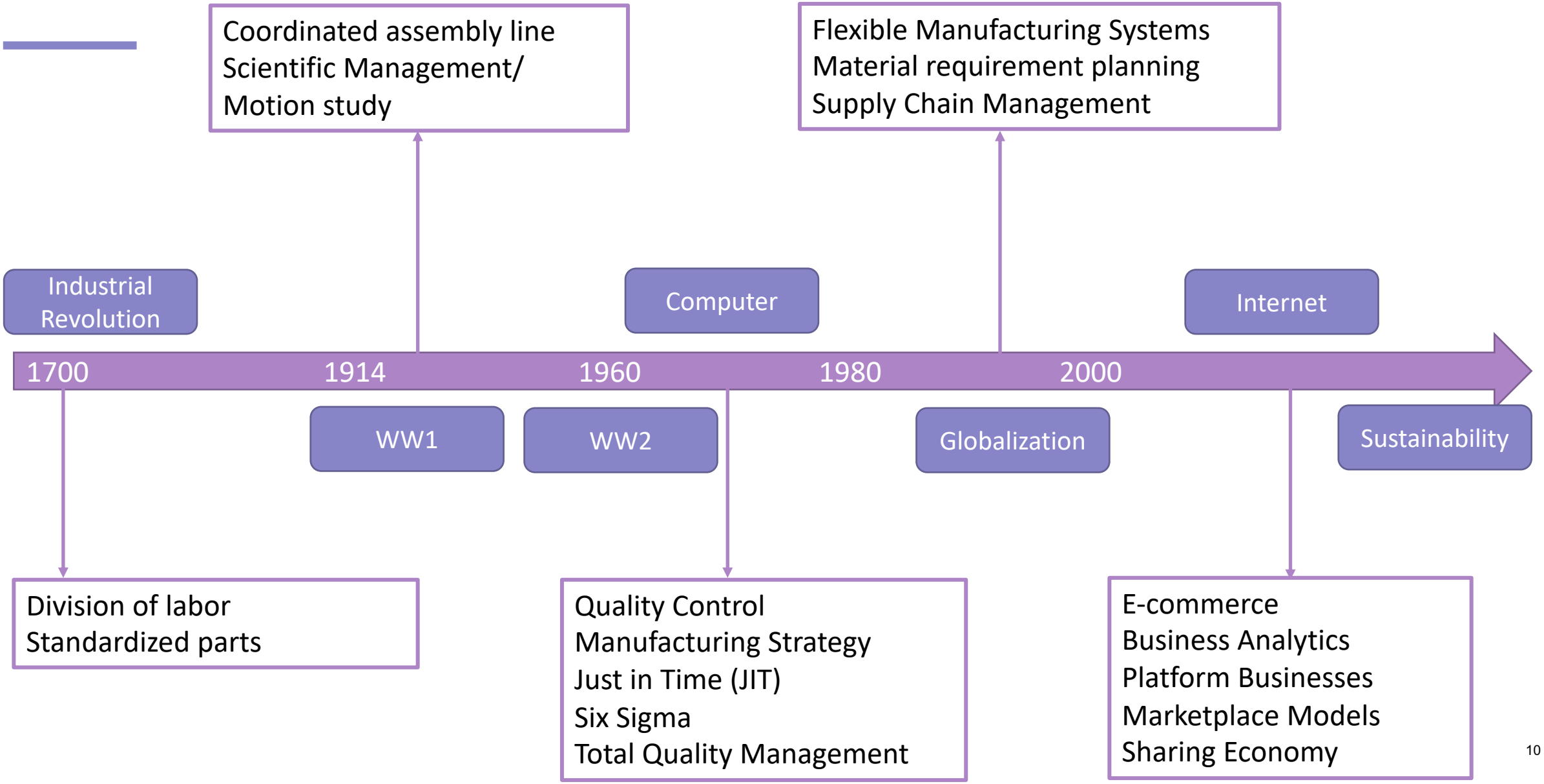
Dimensions of Performance and Trade-offs



Cachon, G., & Terwiesch, C. (2013). *Matching Supply With Demand: An Introduction To Operations Management*. McGraw Hill

Evolution of OM



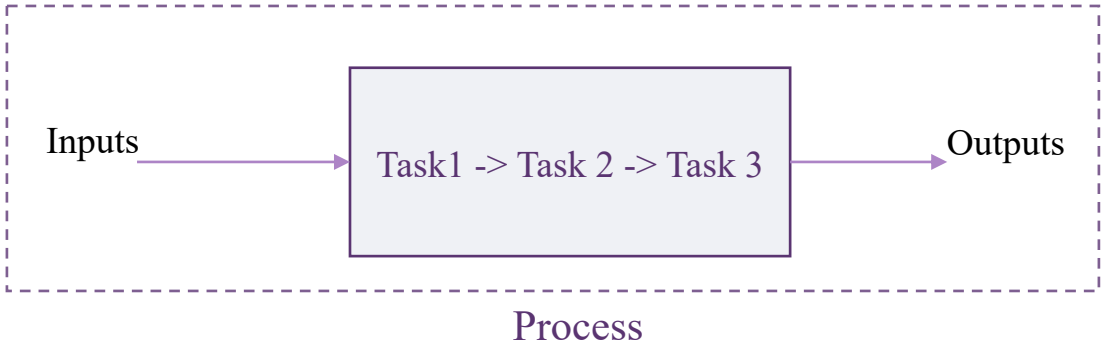
Session 1 and 2: Agenda

- Introduction to OM
- **Process Analysis**
  - Process view
  - Creating a process flow diagram
  - Process performance measures
  - Little’s law
  - Bottleneck
  - Capacity Analysis
  - Product Process Matrix

Operating Process

Process

- A process is a set of tasks to be performed in a defined sequence and uses inputs to create outputs of greater value.
- A building block of operating systems.
- Every organization organizes its work through processes.



Mass manufacturing

Lean manufacturing

Continuous improvement; JIT

Six Sigma

Fast delivery; Logistics network

A Process View- Elements of a process

Example 1: Automobile Manufacturing Process

Process view: A network of **activities** performed by **resources** that transform **inputs** to **outputs**.

Resources: Capital, Labor, Machines

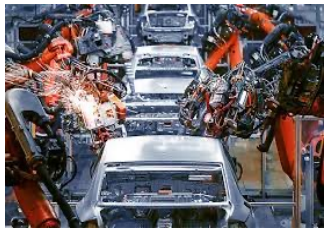
Inputs

Any tangible/ intangible items that flow into the process from the environment

Outputs

Any tangible/ intangible items that flows from the process back to the environment

Flow units : The item being analyzed. It can be specified in terms of inputs or outputs.



Welding --------> Assembly ---------> Painting --------> Inspection

Activities/Sub-processes & Buffer

Activity: Simplest form of transformation

Buffer: stores flow units that have finished one activity but are awaiting another activity.

A Process View- Elements of a process

Process view: A network of **activities** performed by **resources** that transform **inputs** to **outputs**.

Example 2: Treatment in a hospital

Resources: Medical Staff, Beds, Medical equipment

Input

Output

Admission ----------> Treatment ----------> Discharge

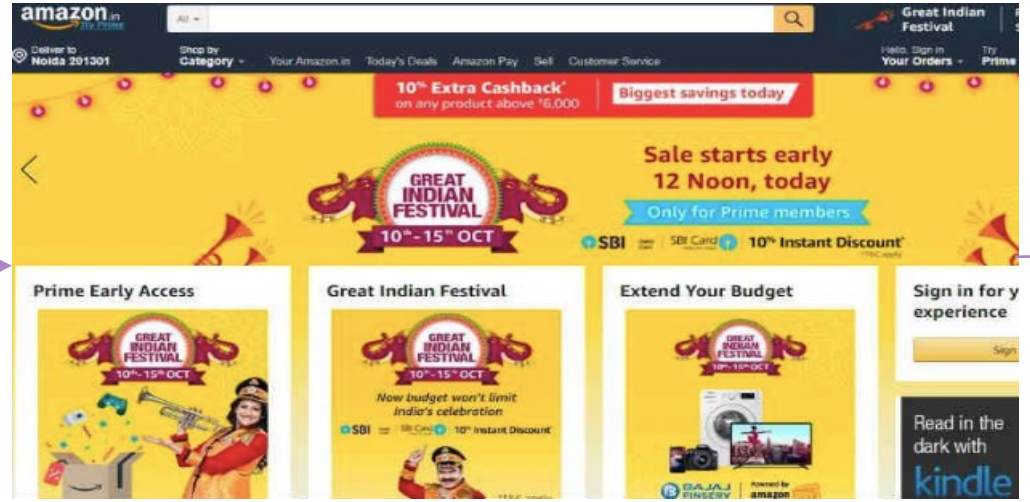
Activities/Sub-processes & Buffer

Process view of order fulfillment (order– delivery)

Example 3: Order-Receipt in an E-commerce website

Resources: Information technology/website, Warehouses, Suppliers, Logistics Network

Customer order



Fulfilled customer order

Order placement ----> Sourcing -----> Packaging ---->Delivery

Process Analysis

Analysis of a process is a key tool thats helps in making operational decisions:

- Make-buy decisions
- Workforce management decisions
- Planning and scheduling

Clearly define the purpose of the analysis- Solve a problem or better understand a problem.

Steps:

1. Create the process flow diagram
2. Use appropriate metrics- to measure and improve

Process flow diagram

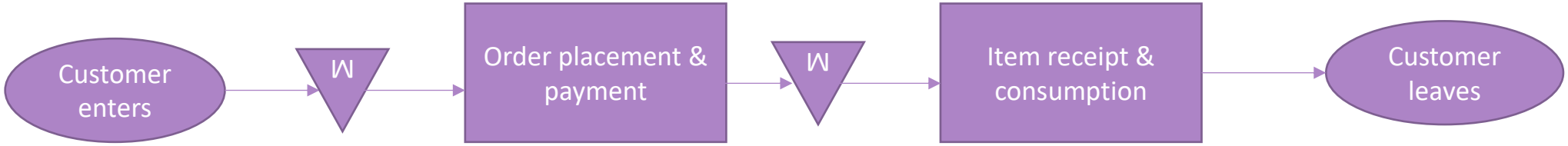
- The first step to understand a process is to describe the process using Process flow diagram or process mapping

- Process flow diagram:

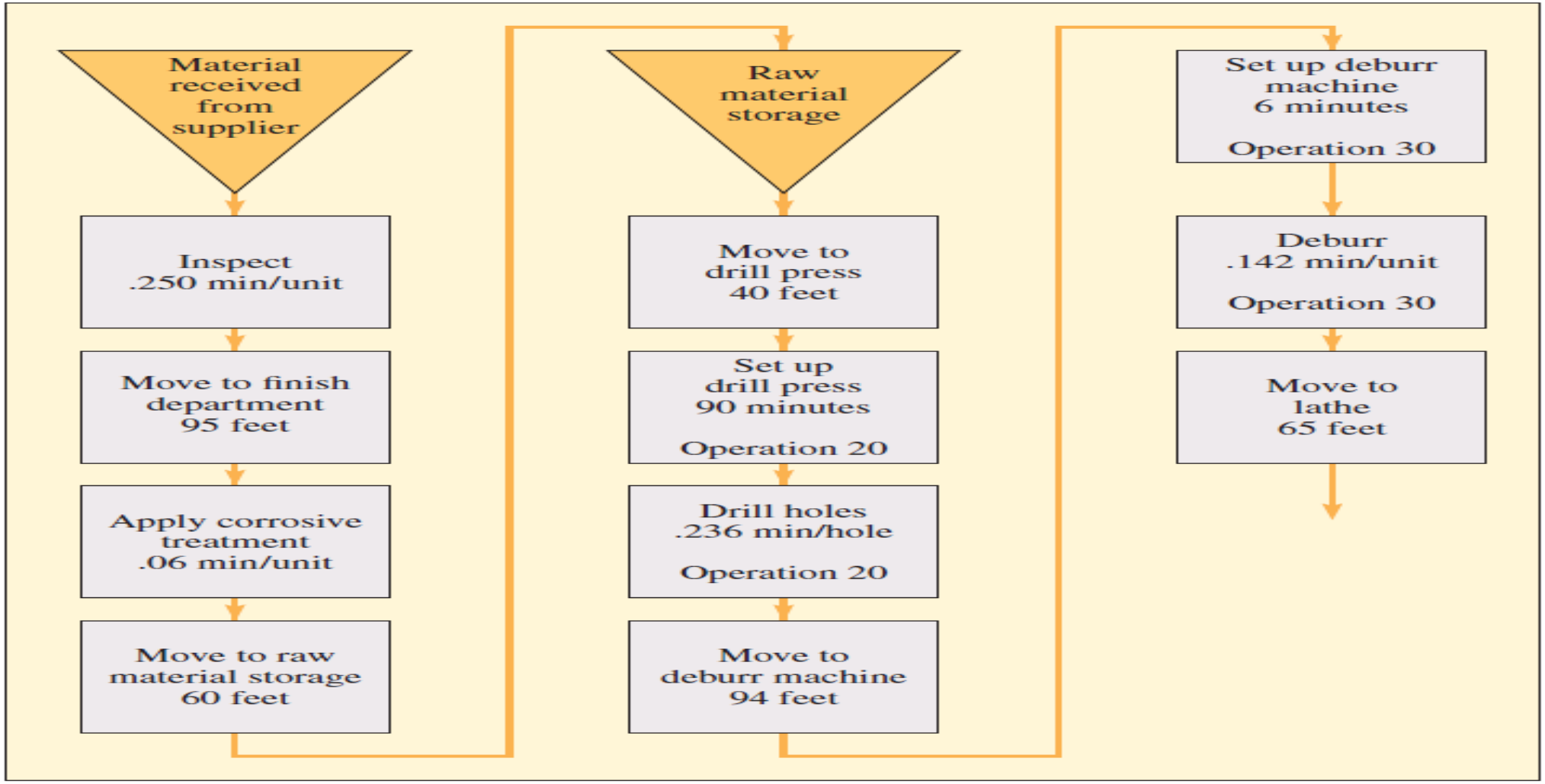
  - A visual representation of the basic elements of a process- activities, flows, and storage areas.

  - Depicts how product/ information progresses through the process.

  - Convenient way to describe a process- a common view of work



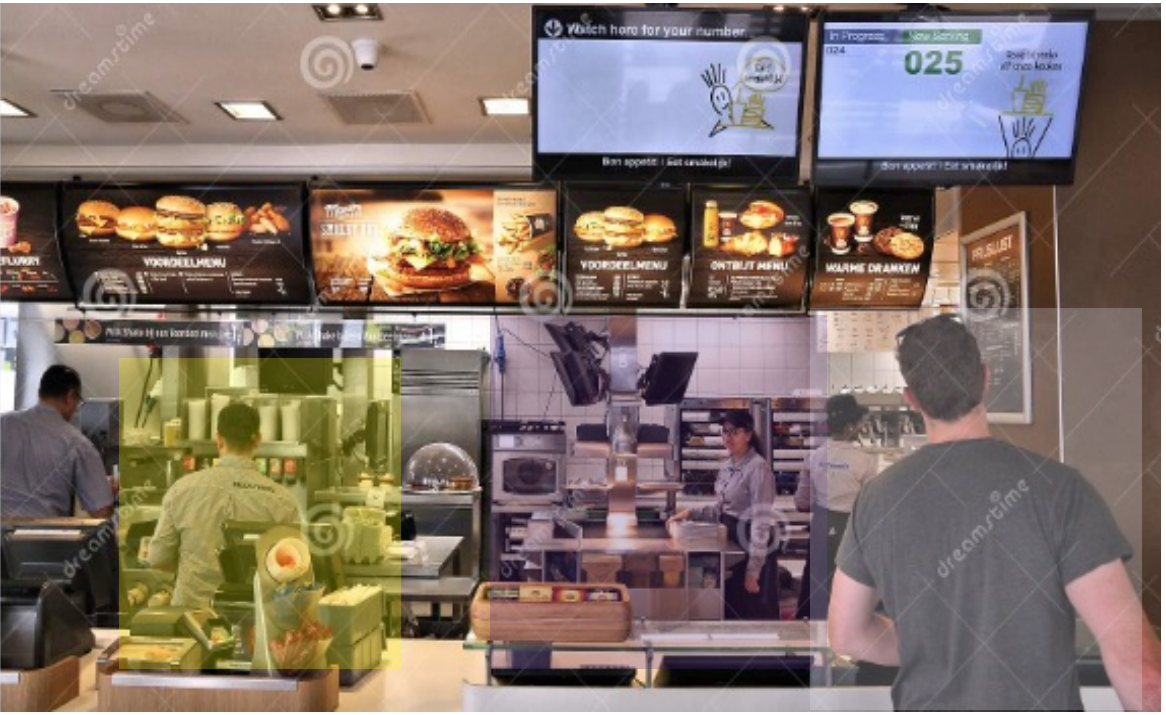
Sample Flowchart

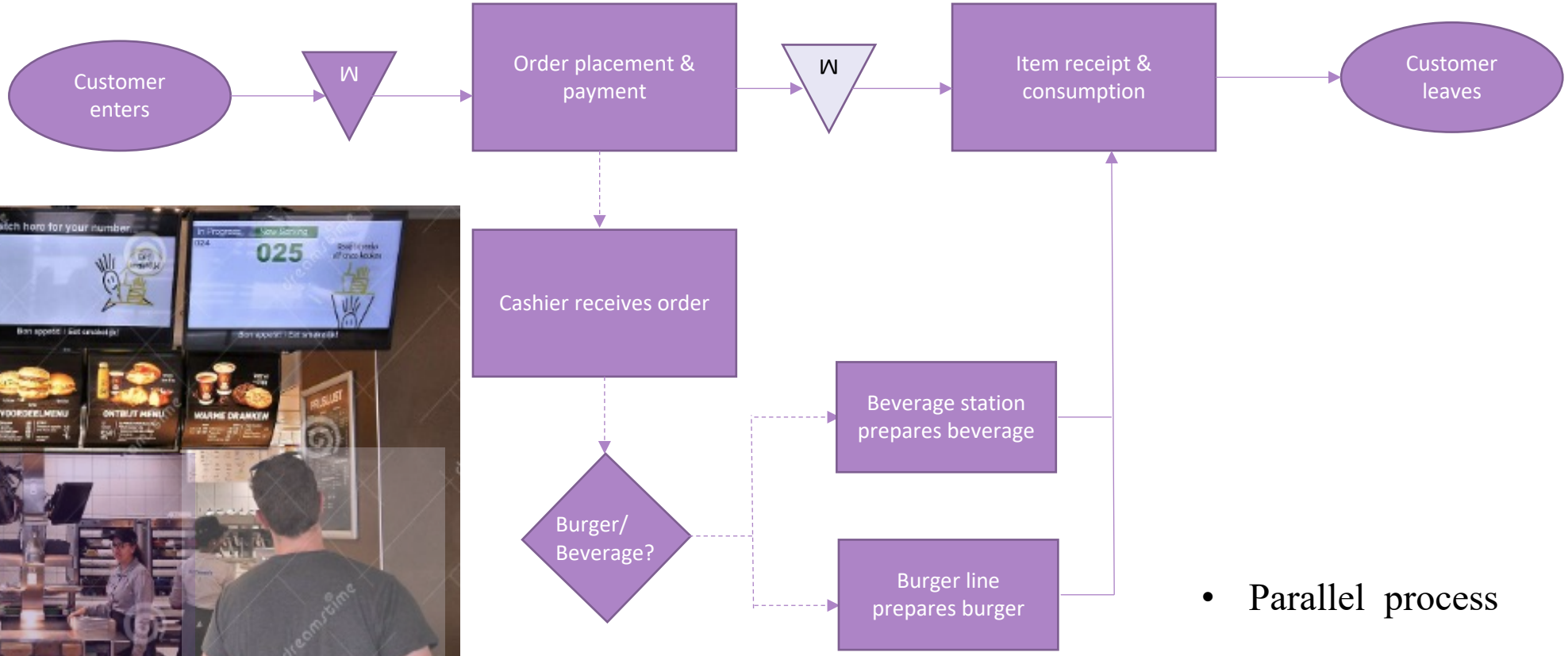


Source: Operations and Supply Chain Management, 14e, Jacobs and Chase

Process Flow diagram

Customers order burger and/or beverage. Orders for beverage are prepared at a dedicated beverage station whereas orders for a burger are prepared in the assembly line format.



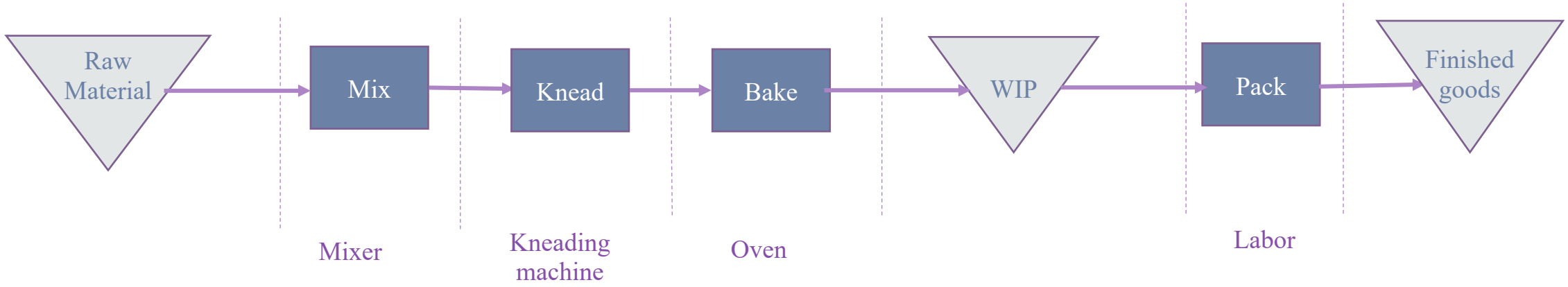


- Parallel process
- Information flows
- Decision node

Process Analysis- Example

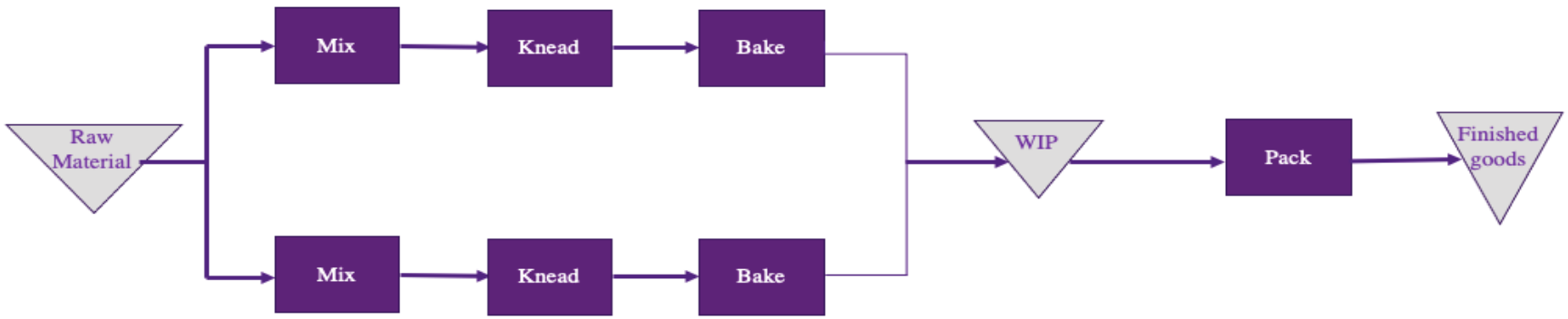
Production of multi-grain bread.

Process steps for preparing multi grain bread	Sequence	Activities/Buffer	Resource
Stock processed wheat, yeast, sugar, salt, milk, other grains, seeds, oil, water	Start	Buffer	-
Mix the ingredients	After all ingredients are available	Mixing	Mixer
Knead the dough	After mixing the ingredients	Kneading	Kneading Machine
Bake the bread	After kneading the dough	Baking	Oven
Pack	After all previous steps	Packing	Labor
Store in dispatch area	End	Buffer	-

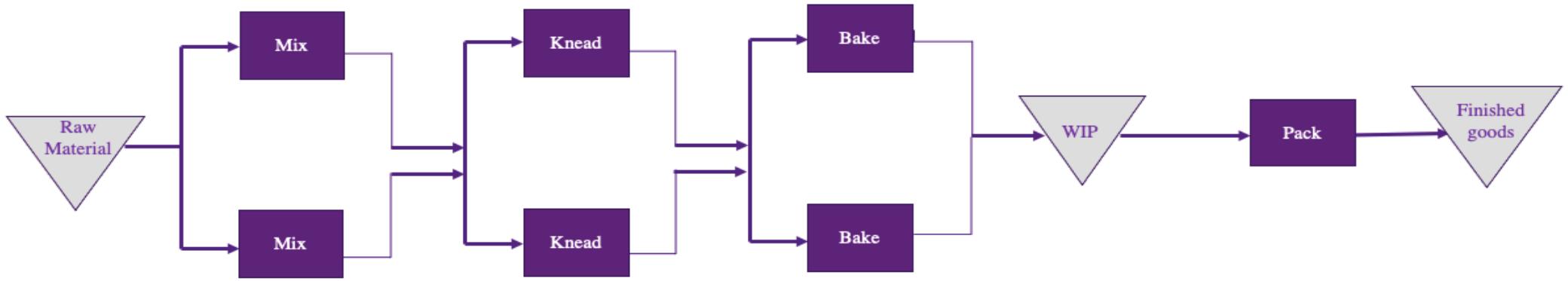


Process flow diagram for parallel lines and parallel activities

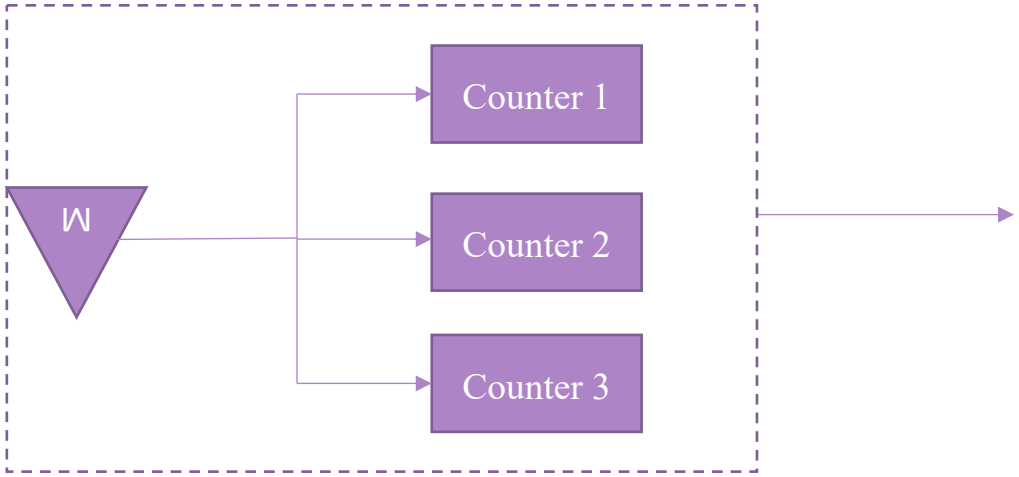
Process flow diagram with two separate lines for Mixing, Kneading and Baking activities:



Process flow diagram with some of the individual activities in parallel:

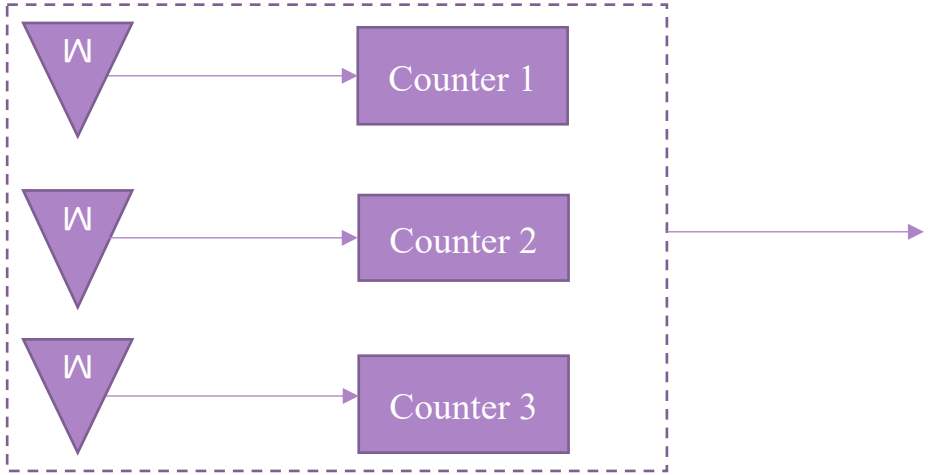


Different types of parallel processing



Single buffer for all parallel tasks

E.g. there is a single queue for check-in luggage at an airline counter. The passengers are directed from the queue to the staff as and when the staff becomes available.



Each parallel task has a dedicated buffer

E.g. there is a separate queue for each billing counter at a supermarket.

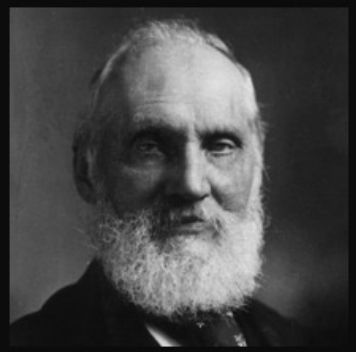
“You cannot manage what you cannot measure.”

– Peter Drucker

“In god we trust. All others must bring data.”

– W. Edwards Deming

Lord Kelvin



When you can measure what you are speaking about, and express it in numbers, you know something about it.

AZ QUOTES

Process Measures/Metrics

Understanding how processes work is essential to ensuring the competitiveness of a company.

Metrics indicate

- How responsive a process is
- How productively a process is operating currently

Key process-flow measures:

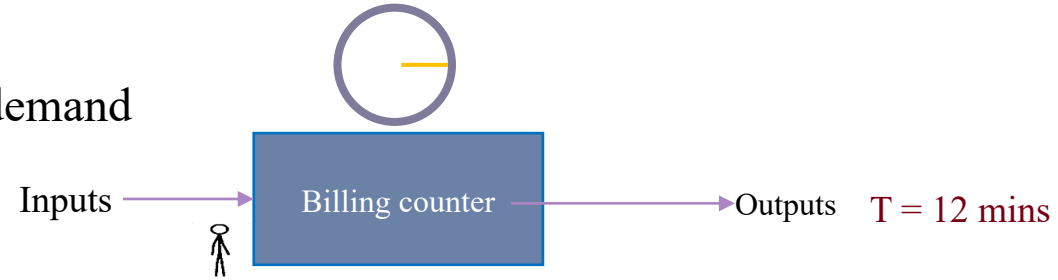
- Flow time
- Flow rate
- Inventory

Together capture the essence of process flow

These measures affect process cost and time

Flow time (T)

- The time it takes for a specific item, job or an order to go through the entire process.
- Time spent by a flow unit within process boundaries
- It indicates the time needed to convert input to outputs
- Provides a sense of how responsive the process is to customer demand



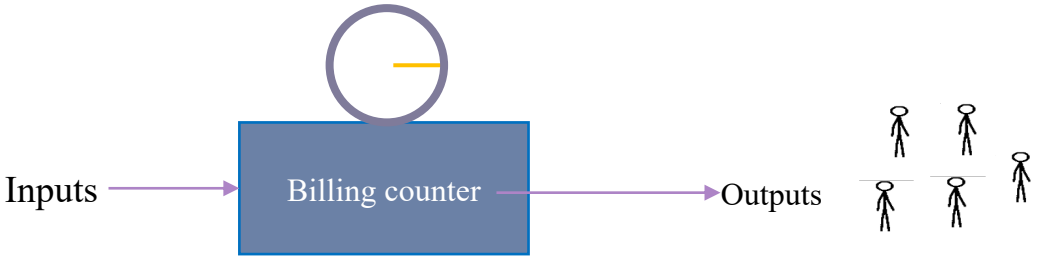
- Useful information to promise a delivery date
- Identifying and reducing the time it takes a customer to pass through a process allows a firm to charge a premium

- How much time will a patient spend in the hospital OPD?
- How long does it take to produce a car?

Flow rate (R)

- The rate at which units flow though a process.

- Average number of flow units that flow through a specific point in the process per unit time.



R= 5 customers /hour

- The flow rate must be aligned with demand to ensure customer needs are met and we are not overproducing.

  - How many patients can the OPD handle per hour?

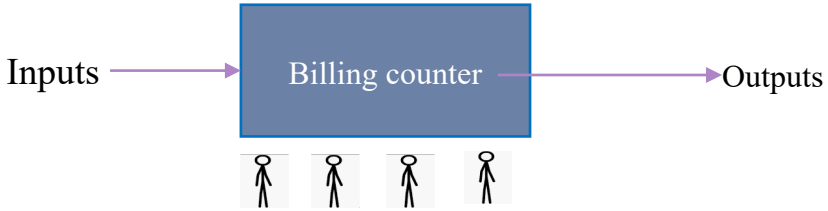
  - How many cars can the factory produce in an 8-hr shift?

Inventory (I)

- Inventory represents the total number of flow units within the process boundary – whether they are being processed or waiting to be processed.

- It can be thought of as a snapshot of inventory

- When the inflow rate exceeds the outflow rate, the number of flow units present within the process increases



I = 4 customers

- At a given time, how many patients are in the OPD of the hospital?

- At a given time, how many cars are in the production stage in the factory?

Little's Law

- Expresses relationship between the process measures **for a stable process**

Average Inventory (I) = Throughput (R) x Average Flow time (T)

Implications:

- A manager needs to focus on only two measures out of the three
- For a given throughput, the only way to reduce flow time is to reduce inventory

Application of Little's Law

Two-wheeler production factory:

It takes 12 hours to make a two-wheeler

The plant assembles 200 two-wheelers per 8-hour shift

The plant operates one shift

Calculate the average number of two-wheelers under production in the factory.

Average Inventory

- Average Inventory= Throughput x Flow time
- Average Inventory = 25 batteries per hour x 12 hours
- Average Inventory = 300 batteries

Application of Little's Law

2) A grocery store's billing counter processes 30 customers per hour. On average there are 4 customers in a line. What is the flow time?

Flow rate: R = 30 customers/hour

Inventory: I = 4 customers

Flow time: $T = \frac{I}{R} = \frac{4}{30} hours = \frac{4 * 60}{30} mins => T = 8\ mins$

A customer joining the billing line is expected to wait on average 8 mins to be processed and leave the store.

3) A bank processes 5000 checks in a week. Average processing time for a check is 15 mins. Assume the bank works for 10 hrs a day for 5 days per week. How many checks are there in the bank at any given time?

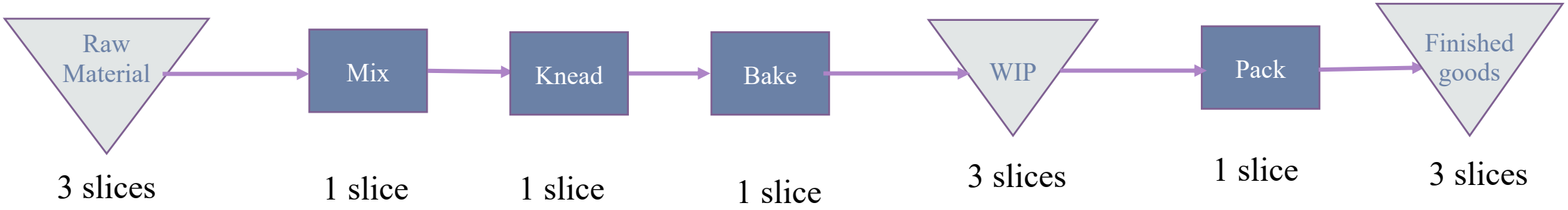
R= 5000 checks/wk

T= 15 mins = $\frac{15}{5*10*60} = \frac{1}{200}$ weeks

$I = T * R = 5000\ checks/wk * \frac{1}{200} wk = 25\ checks$

In the branch, on an average, there are 25 checks in various stages of processing.

Application of Little's Law



The average throughput of the bread making process is given- 120 slices per hour. Assume there is one slice of bread in process at each activity stage and three slices of bread at each storage point. Compute the average flow time.

R= 120 slices/hour

$$I = 3 + 1 + 1 + 1 + 3 + 1 + 3 = 13\ slices$$

T= $\frac{I}{R} = \frac{13}{120}$ hours = $\frac{13 * 60}{120}$ minutes = 6.5 minutes

A slice of bread spends around 6.5 minutes in various stages of the bread-making and packaging process.

Capacity

A critical feature of any system is its ability to produce/process enough to meet demand

- Capacity of a resource: Maximum flow rate if the resource were fully utilized
- Capacity of a process:
    - The capacity of a process is the capacity of its limiting resource.
    - It denotes the maximum output that the process can provide in a given time.
    - It helps to understand the extent to which the demand can be met.
    - It is equal to the capacity of the bottleneck resource i.e. the smallest resource capacity.
    - Bottleneck: slowest resource of a process

Utilization

- Degree to which resources are utilized by a process
- Utilization $of a\ resource = \frac{\text{actual flow rate of the resource}}{\text{the maximum flow rate of the resource}}$

E.g.

1. A flight can carry 200 passengers. However, only 150 bookings are made.

$$\text{Utilization } of\ the\ flight's\ seating\ capacity = \frac{150}{200} = 0.75 \text{ or } 75\%$$

2. A denim manufacturing factory has a maximum capacity to produce 10000 denims/month.
   Due to the pandemic, the demand forecast is around 6500 units/month for the next few months.

Utilization of the factory = 6500/10000 = 0.65 or 65%

However, in the festive season the demand is 12000 units/month.

$$\text{Utilization } of\ the\ factory = \frac{10000}{10000} = 1 \text{ or } 100\%$$

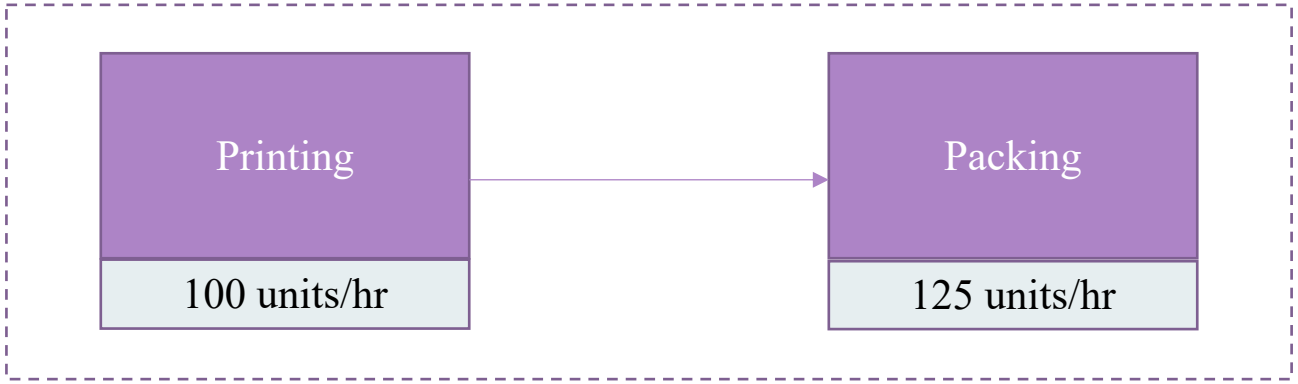
Bottleneck

Bottleneck: slowest resource of a process

Bottleneck limits the production/processing rate

Capacity of a process is equal to the capacity of the bottleneck resource i.e. the resource with smallest capacity.

Capacity Analysis

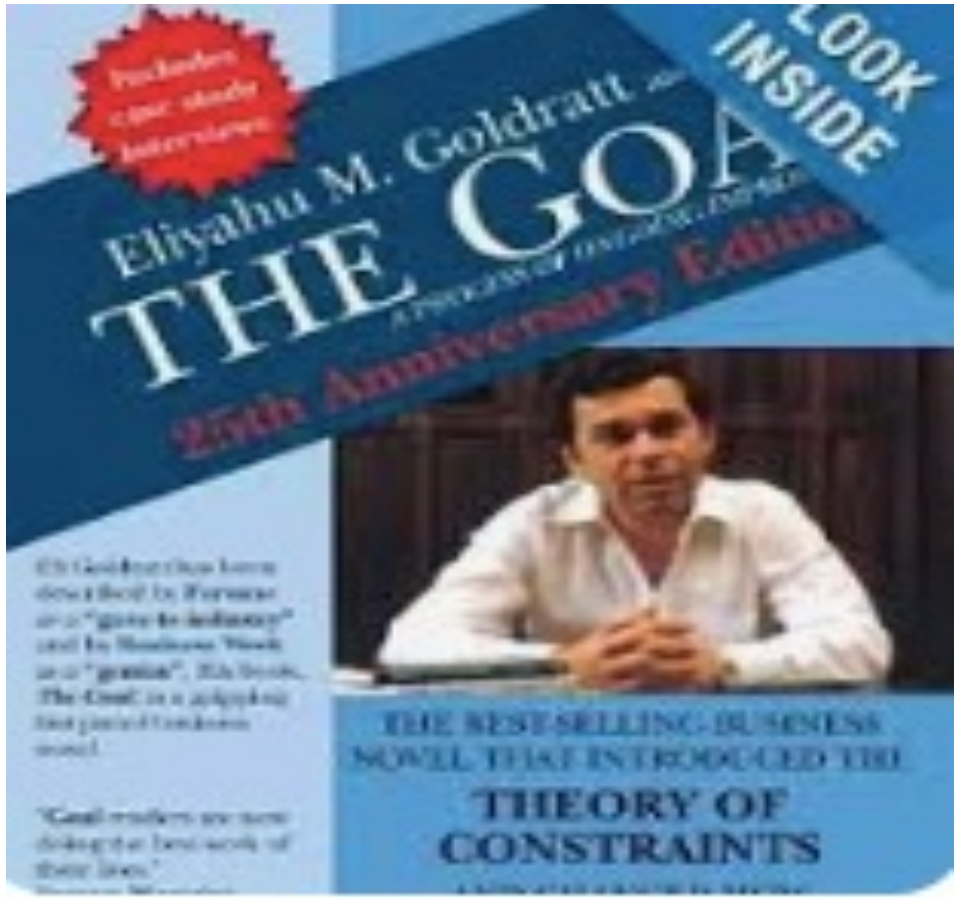


Activity	Resource	No. of resource units	Max throughput (Units/hr)	Process Capacity (Units/hr)	Resource utilization (%) 100*Process capacity/Resource Capacity
Printing	Worker	1	100	Min(100,125) = 100	100%
Packing	Worker	1	125		80%

Process Capacity = 100 units/hr

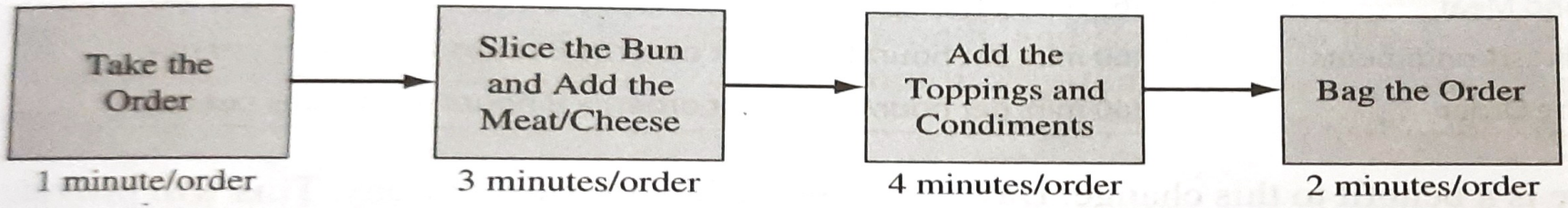
Theory of Constraints

1. In a multistage process, identify the bottleneck – is it inside the system or outside the system (market demand)
2. Bottleneck is the weakest link in the system
3. Adding resource at a non-bottleneck resource is a mirage
4. Improve the performance of the bottleneck
5. Check where the bottleneck has shifted



Process analysis example:

Daffy Dave's Sub Shop makes custom submarine sandwiches to order. It is analyzing the processes at its shop. The general flow of the process is shown as follows. A different person is working at each of the steps in the process.



Daffy Dave wants to figure out the following for a typical eight-hour work day.

a. What is the current maximum output of the process?

b. If we add another person, where would we add him or her and what is the benefit?

c. Is there a benefit if we can shift one minute from Bun and Meat to Order Taking? Assume we do not make the change in part (*b*).

d. Is there a benefit if we shift one minute of work from Condiments to Bagging? Assume we do not make the changes in parts (*b*) and (*c*).

Source: Operations and Supply Chain Management, 14e, Jacobs and Chase

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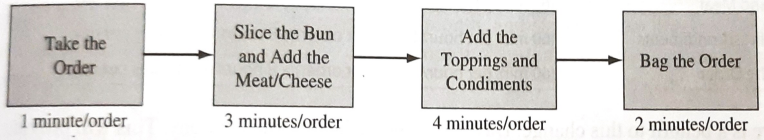

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a. In a day, output of each activity:

1. 1 min/order = > 1 order/ min = 60 orders / hour = 60*8 = 480 orders/day
2. 3 min/order=> 0.33 orders/min = 20 orders/hour = 20*8 = 160 orders/day
3. 4 min/order = > 0.25 order/ min = 15 orders / hour = 15*8 = 120 orders/day
4. 2 min/order=> 0.5 order/ min = 30 orders / hour = 30*8 = 240 orders/day

Maximum output of the entire process is restricted by the slowest activity. Hence, maximum output of the shop is 120 orders/day.

b. Increasing capacity i.e., adding resource for any activity other than bottleneck is a mirage.
The maximum output or the process capacity is limited by the bottleneck.

- If station 1, 2 or 4 improve, station 3 will still remain the bottleneck.
- Hence, the shop can improve the maximum output only if an additional resource is added where bottleneck is i.e. station 3.

c. In a day, output of each activity:

1. 2 min/order = > 0.5 order/ min = 30 orders / hour = 30*8 = 240 orders/day
2. 2 min/order=> 0.5 order/ min = 30 orders / hour = 30*8 = 240 orders/day
3. 4 min/order = > 0.25 order/ min = 15 orders / hour = 15*8 = 120 orders/day
4. 2 min/order=> 0.5 order/ min = 30 orders / hour = 30*8 = 240 orders/day

No benefit as the bottleneck remains unchanged.

d. In a day, output of each activity:

1. 1 min/order = > 1 order/ min = 60 orders / hour = 60*8 = 480 orders/day
2. 3 min/order=> 0.33 orders/min = 20 orders/hour = 20*8 = 160 orders/day
3. 3 min/order=> 0.33 orders/min = 20 orders/hour = 20*8 = 160 orders/day
4. 3 min/order=> 0.33 orders/min = 20 orders/hour = 20*8 = 160 orders/day

There is a benefit as the shop can now serve 160 orders/day.

Source: Operations and Supply Chain Management, 14e, Jacobs and Chase

Customer order decoupling point

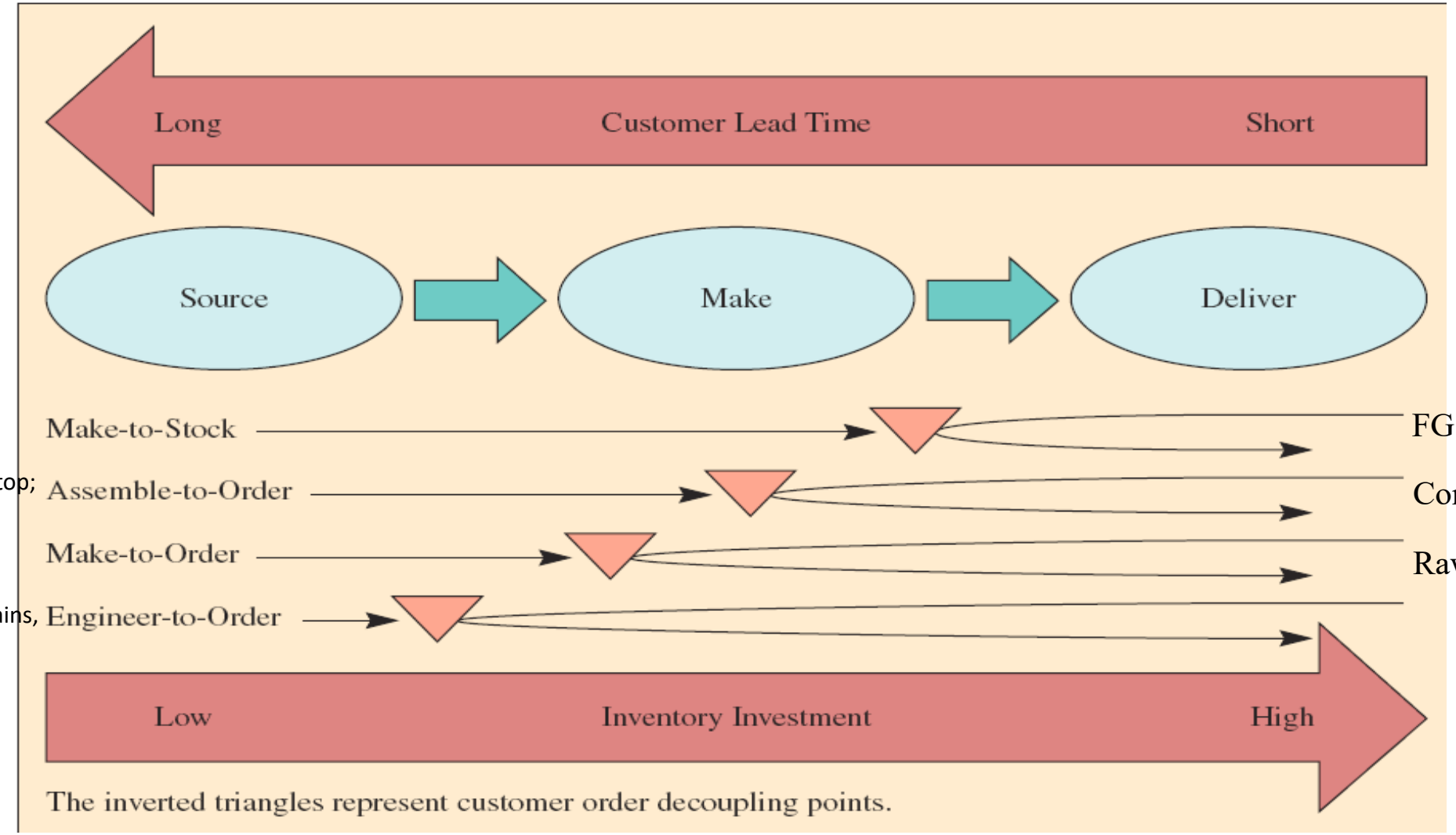
Inventory helps to maintain independence of operations

Most FMCG and electronic items

Electronic items e.g. Dell laptop; Subway sandwiches

Tailored clothes

Construction of airplanes, trains, real estate projects



FG inv

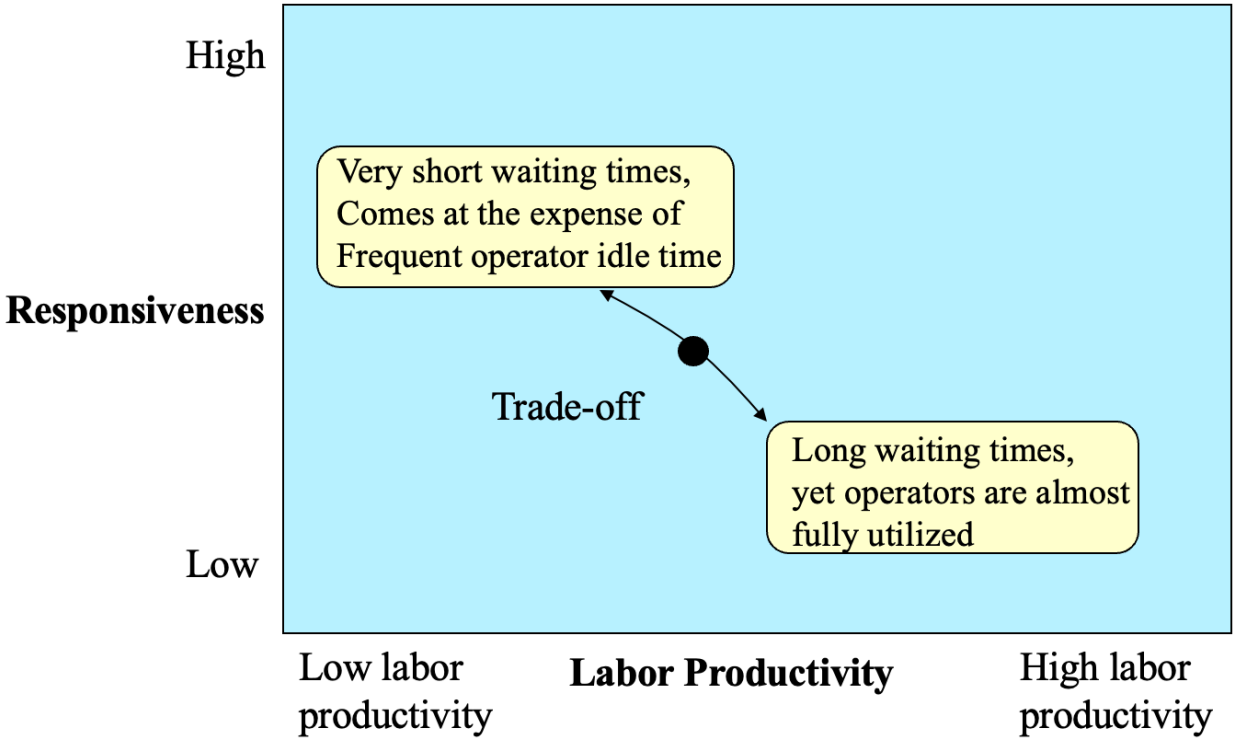
Components inv

Raw materials in

Source: Operations and Supply Chain Management, 14e, Jacobs and Chase

OM provides tool for strategic trade-offs

A firm often faces a trade off between improving labor productivity (utilization of labor) and being responsive. Responsiveness is defined as the ability to respond to changes in market demand.



If the utilization levels are very high, this will give less time to the system to quickly respond to the changes in market demand, thereby lowering its responsiveness.

On the other hand, if a company is very responsive to customer demand, the company will need to be sensitive to the change in demand pattern and keep some idle capacity.

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